Figure 5: Initiating hob tool

Figure 7: Hob selection page where you can select base image type and customise layers

Let's try Hob here—see Figure 5 (terminal) and Figure 6 (Hob GUI start-up). The next step is shown in Figure 7. As you can see, the target *qemux86* has been selected. Next, select *'core-image-minimal'* and kick off the build. Hob starts building (Figure 8). This is going to take some time, depending on the config and parallelism options you have set.

After the build, you can execute *runqemu qemux86* to run the Qemu emulator with the newly compiled image.

Some of the other configuration files in the Poky system are:

1) Configuration (**.conf*)—global variables
- *build/conf/local.conf* (local user-defined variables)
- *./meta-yocto/conf/distro/poky.conf* (Yocto policy config variables)
- *./meta/conf/machine* (machine-specific variables)

2) *conf/local.conf*—additional features:
- *EXTRA_IMAGE_FEATURES* adds features (groups of packages)
- *INCOMPATIBLE_LICENSE* = *'GPLv3'* eliminates packages using this licence

3) Recipes for building packages
- For instance, *meta/recipes-core/glib-2.0/glib-2.0_2.32.4.bb* builds GLib and installs it.

The documentation provides excellent explanations for many other files.

You must, however, watch out for the following:
- If custom patches, which are pulled from upstream, are incorporated in the source, Yocto needs to carry them within, and apply and fix these patches for every future update.
- Kickstarting a project in Yocto involves a fair bit of set-up work initially.
- Yocto does not mandate any version control while pulling code from upstream.

Figure 6: Hob screenshot while it is parsing various recipients

Figure 8: Hob build process which will produce the final image

Getting involved

Like every other open source project out there, Yocto depends on contributions from members like you and me. Refer to the contribution pages at *yoctoproject.org*, and be a part of this excellent project.

Mastering Yocto tools effectively can be a little daunting initially, but the time spent on it is well worth it. The flexible customisation and configurable options give versatile project-specific control, which will help developers of embedded systems to concentrate on their core software development. END

References

[1] *http://www.yoctoproject.org*
[2] *#yocto on irc freenode.irc.net (#yocto channel)*

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